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Wiley
International Journal of Intelligent Systems
Volume 2024, Article ID 6528752, 18 pages
https://doi.org/10.1155/int/6528752

*Research Article*

# Enhancing Breast Cancer Detection in Ultrasound Images: An Innovative Approach Using Progressive Fine-Tuning of Vision Transformer Models

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Received 3 June 2024; Accepted 7 November 2024

Academic Editor: Eugenio Vocaturo

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Breast cancer is ranked as the second most common cancer among women globally, highlighting the critical need for precise and early detection methods. Our research introduces a novel approach for classifying benign and malignant breast ultrasound images. We leverage advanced deep learning methodologies, mainly focusing on the vision transformer (ViT) model. Our method distinctively features progressive fine-tuning, a tailored process that incrementally adapts the model to the nuances of breast tissue classification. Ultrasound imaging was chosen for its distinct benefits in medical diagnostics. This modality is noninvasive and cost-effective and demonstrates enhanced specificity, especially in dense breast tissues where traditional methods may struggle. Such characteristics make it an ideal choice for the sensitive task of breast cancer detection. Our extensive experiments utilized the breast ultrasound images dataset, comprising 780 images of both benign and malignant breast tissues. The dataset underwent a comprehensive analysis using several pretrained deep learning models, including VGG16, VGG19, DenseNet121, Inception, ResNet152V2, DenseNet169, DenseNet201, and the ViT. The results presented were achieved without employing data augmentation techniques. The ViT model demonstrated robust accuracy and generalization capabilities with the original dataset size, which consisted of 637 images. Each model's performance was meticulously evaluated through a robust 10-fold cross-validation technique, ensuring a thorough and unbiased comparison. Our findings are significant, demonstrating that the progressive fine-tuning substantially enhances the ViT model's capability. This resulted in a remarkable accuracy of 94.49% and an AUC score of 0.921, significantly higher than models without fine-tuning. These results affirm the efficacy of the ViT model and highlight the transformative potential of integrating progressive fine-tuning with transformer models in medical image classification tasks. The study solidifies the role of such advanced methodologies in improving early breast cancer detection and diagnosis, especially when coupled with the unique advantages of ultrasound imaging.

**Keywords:** breast cancer; deep learning; fine-tuning; medical image classification; transformer models; vision transformer (ViT)

## 1. Introduction

Breast cancer remains one of the most significant health challenges facing the medical community, impacting millions of women worldwide. According to the World Health Organization, breast cancer remains the most common cancer among women globally and the second most common cancer overall, accounting for over 2.1 million diagnoses annually. The critical nature of early detection and diagnosis in the management of breast cancer cannot be

overstated, often serving as the pivotal factor that determines patient outcomes [1, 2]. The use of imaging techniques, particularly ultrasound, plays a vital role in the early detection strategies of breast cancer. Ultrasound imaging is favored for its noninvasive nature and the ability to provide real-time insights into the breast tissue, enabling the differentiation of benign and malignant tumors [3]. Despite these advantages, the interpretation of ultrasound images presents considerable challenges, primarily due to the nuanced differences between tumor types. Such challenges have prompted increased research into computational methods to augment the diagnostic process [4].

Machine learning (ML) and deep learning (DL) have emerged as frontrunners in the computational approach, with DL models showing exceptional prowess in image classification tasks. These advanced models, powered by their ability to learn from extensive datasets, have consistently demonstrated superior performance over traditional ML techniques in various domains, including object recognition, facial recognition, and medical image analysis. DL models have been instrumental in classifying images of different tumor types with promising accuracy in medical imaging. This has opened avenues for leveraging these models to refine the diagnostic accuracy for breast cancer, potentially revolutionizing the early detection process and improving prognostic outcomes. The authors of [5] developed a predictive model for patients' overall 5-year survival with breast cancer by employing four feature selection techniques, ultimately creating a Bayesian dynamic Cox (BDCox) model to enhance model parsimony. This approach is noteworthy for its focus on survival prediction, which contrasts with our emphasis on image classification. The authors of [6] improved physician image ratings by utilizing cancer detection computer programs, specifically developing a 3D convolutional neural network (CNN) to analyze ABUS images and identify suspicious areas. This method's emphasis on minimizing false positives aligns with our goal of enhancing diagnostic accuracy, although our work focuses on employing the ViT model to distinguish between benign and malignant tumors in ultrasound images.

The task of detecting breast cancer from ultrasound images is uniquely challenging due to the inherent complexity of medical imaging. These images are often riddled with noise, speckles, and highly complex tissue textures, making it a daunting task to extract meaningful features using traditional ML methods. To tackle these challenges head-on, we turn to the Vision Transformer (ViT) model, leveraging its unique strengths in capturing long-range dependencies and modeling complex spatial relationships. This section outlines the motivations behind selecting the ViT model and the progressive fine-tuning approach while comparing its benefits to traditional CNNs.

Traditional CNNs are adept at detecting localized patterns within an image using convolutional filters operating over small patches. However, their effectiveness is limited when it comes to capturing long-range dependencies across the image, a crucial aspect of medical imaging. The ViT introduces a game-changing approach by using a self-attention mechanism that enables the model to capture relationships between distant regions of the image. This is particularly beneficial in ultrasound imaging, where both local textures and global patterns may contain vital diagnostic information.

In ultrasound images, noise and artifacts such as speckles add complexity to the task of breast cancer detection. The ViT's self-attention mechanism is a game-changer, enabling the model to weigh the importance of different regions of the image. This unique feature allows the model to efficiently focus on both fine details and larger structures simultaneously, giving the ViT a significant advantage in detecting subtle anomalies that may be distributed across different parts of the image. By modeling both local and global spatial relationships, ViT enhances the ability to accurately classify breast tissue as benign or malignant.

Another key strength of the proposed approach is the use of progressive fine-tuning. The ViT model is initially pretrained on large datasets of general images, learning broad features applicable across multiple domains. We then progressively fine-tune the model on breast ultrasound images, allowing it to focus on features specific to breast cancer. This gradual approach helps the model specialize in the nuances of breast cancer detection, such as identifying patterns that distinguish benign from malignant lesions.

Progressive fine-tuning improves model accuracy and reduces the risk of overfitting, a common issue when training DL models on smaller datasets. By starting with a general understanding of image features and progressively narrowing its focus, the model can adapt more effectively to the specific characteristics of breast ultrasound images. This method also ensures that the model can be generalized to unseen data, making it more robust for clinical applications.

While CNNs have been the dominant model for medical image analysis, they possess limitations when applied to more complex tasks like breast cancer detection in ultrasound images. CNNs primarily operate using convolutional filters that capture local features but struggle to model long-range dependencies across the entire image. In contrast, the ViT model's self-attention mechanism is designed to learn local and global image structures, making it more effective for tasks where subtle patterns might appear in different image regions.

The ability of ViT to learn holistic representations of the image is crucial in breast ultrasound imaging, where both fine textures and larger anatomical structures may play a role in diagnosis. CNNs often require deeper architectures to achieve a similar feature extraction level, which increases computational costs and training time. ViT, by comparison, is more efficient in capturing essential features, offering a more advanced and scalable solution for breast cancer detection.

Our study delves into the application of the ViT model. This transformer-based DL model has garnered attention for its remarkable performance in several image recognition tasks. Unlike conventional models that rely on convolutional layers, the ViT model leverages self-attention mechanisms, which enable it to learn spatial hierarchies from data. This feature is particularly beneficial in medical imaging, where the spatial relationship between pixels can denote critical

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diagnostic information. We propose a novel application of the ViT model for breast cancer detection in ultrasound images, utilizing a technique known as progressive fine-tuning. This method involves incrementally training the model on the classification task, allowing the model to adapt progressively. The fine-tuning starts with the final layers of the model, which are typically more specialized for the task and proceed to earlier layers that encapsulate more general features. Our approach is designed to tailor the ViT model's robust feature extraction capabilities to the nuances of breast ultrasound images. This paper presents a comprehensive investigation into the efficacy of the ViT model, enhanced by progressive fine-tuning, for classifying benign and malignant breast ultrasound images. Our goal is to advance the methodologies in early breast cancer detection and diagnostic precision, striving to improve the current standards and offer a new benchmark in this pervasive disease's early and accurate diagnosis. Through this research, we aim to contribute a significant piece to the puzzle of breast cancer management, offering hope for enhanced patient outcomes through the power of AI-driven diagnostics.

The paper is structured as follows: the related work in Section 2 provides a review of existing literature on breast cancer detection using DL methodologies. The methodology is proposed in Section 3 and is subdivided into dataset description, pretrained DL models selection, ViT model explanation, progressive fine-tuning process elaboration, fine-tuning classification of the ViT model, and methodology evaluation. The results are presented in Section 4, which presents findings from Experiment 1 focusing on pretrained DL models, Experiment 2 emphasizing progressive fine-tuning ViT, and Experiment 3 discussing the fine-tuned classification head of ViT model. Finally, the discussion and conclusion are explored in Section 5, which synthesizes findings, underscores the significance of the research, and suggests avenues for future exploration in breast cancer detection.

## 2. Related Work

The use of DL in breast cancer detection has made significant improvements to medical imaging and diagnosis. These powerful algorithms can analyze large amounts of data, learning patterns, and details that might be difficult for traditional methods to identify. By using DL, doctors can now detect breast cancer with greater accuracy and sensitivity at various stages, leading to earlier and more precise diagnoses. This benefits patient outcomes and allows for more personalized treatment plans. As DL research and development continue to advance, integrating these models holds great promise for transforming breast cancer detection and ultimately enhancing the effectiveness of breast cancer screening programs overall.

In selecting the related works for inclusion in this study, we focused on research that directly aligns with our objective of improving breast cancer detection in ultrasound images using advanced DL methodologies, particularly transformer-based models. The criteria for including related works were as follows: relevance to ultrasound imaging, use of DL models, recent advances, and methodological innovation.

The authors of [7] investigated the application of DL methods for precise diagnosis and classification of breast cancer, specifically focusing on Adam's optimization algorithm and a two-layer neural network. The study highlights the optimization of network weights to address common problems during DL training, such as data instability and overfitting. Trials utilizing various learning rates and datasets, such as WBCD, WDBC, and Coimbra, show how well the suggested method performs, as evidenced by its astounding 96% classification accuracy for breast cancer. The authors of [8] involved the application of DL approaches in predicting breast cancer. The researchers developed a new 19-layer deep CNN model to separate the benign and malignant tumors. K-fold cross-validation experiments were carried out on BreaKHis dataset. Then, the proposed CNN model was compared with a support vector machine (SVM) and the GoogLeNet DL model. The results showed that the new 19-layer deep CNN model performed better than the other methods, achieving the highest accuracy with a value of 84.5%, AUC with a value of 85.7%, and very low classification MAE with a value of 0. These observations suggest that the model helps to classify breast cancer and may be used to improve diagnostic accuracy.

The authors of [9] aimed to develop a novel 19-layer deep CNN model for automated breast cancer diagnosis. Using the BreaKHis dataset, the model's capacity to distinguish between benign and malignant tumors will be assessed. In an attempt to demonstrate that their new model is better at predicting breast cancer, the researchers use metrics such as accuracy, AUC, MAE, and MSE to evaluate the performance of their model against that of state-of-the-art DL model (GoogleNet) and traditional ML (SVM). The authors of [10] implemented a novel DL-based method for classifying and detecting breast cancer in mammography images from the CBIS-DDSM dataset. It does this by utilizing the EfficientNet architecture. The accuracy of 75% and the area under the curve (AUC) of 83% achieved by the suggested method show how well DL techniques work to improve medical imaging for the detection of breast cancer. Improving diagnostic capabilities by incorporating these cutting-edge technologies into diagnostic procedures is highly promising, which could result in earlier precise interventions. The authors of [11] aimed to automate the diagnosis process of breast cancer using ML and DL algorithms. The researchers compared various algorithms, including SVM, logistic regression, KNN, random forest, naive Bayes classifier, and ANN, using the Wisconsin Breast Cancer (Diagnostic) dataset. They found that the ANN algorithm achieved the highest accuracy of 98.7%, demonstrating its potential for improving breast cancer detection and classification.

The authors of [12] addressed the problem of early diagnosis and detection of breast cancer by creating a reliable technique for early cancer cell detection from mammography scans. An adaptive deep convolution neural network (ADCNN) is used as a suggested method to identify signs of breast cancer from mammography images. The objective is to use the images to categorize the detected cells as normal,

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benign, or malignant. The principal objective is to enhance early detection of breast cancer and subsequently lower the disease's death rate. The authors of [13] investigated the potential of ML and intense learning methods in improving breast cancer detection and diagnosis. The researchers compared three different models using the BreakHis dataset and achieved an accuracy of 99% for training and 98% for validation. These findings suggest that DL models can improve breast cancer diagnosis and treatment. The authors of [14] proposed a novel ensemble learning-based architecture for breast cancer classification using DL techniques. The proposed architecture combines the predictions of three individual models: Xception, ViT, and XceptionBiLSTM. The results show that the proposed architecture achieves high accuracy (90% in the hidden test dataset, 95% accuracy, and 94.9% $F1$-score in the validation dataset), demonstrating the significant potential of the ensemble learning-based architecture for breast cancer classification. The authors of [15] used transfer learning, CNNs, and the ResNet-50 and Inception v3 models to develop a computer-aided detection system for breast cancer. The objective is to use Inceptionv3 transfer learning to detect breast cancer accurately. To improve the usability and accessibility of the suggested detection system for physicians and patients, a graphical user interface (GUI) will be created. This technology may enhance the efficacy and precision of breast cancer identification, improving patient outcomes.

The authors of [16] improved breast cancer diagnosis and treatment by using advanced ML and DL techniques for automated mammogram analysis. It identifies a limitation in current research, where tumors are simply classified as normal/abnormal or benign/malignant. This study proposes a more detailed approach with three classification categories. The research finds that DL-derived features are superior to handcrafted ones, and transfer learning is preferable over traditional methods when data is scarce. The authors of [17] analyzed medical images using DL radiology (DLR), and hidden details in this study allowed doctors to dynamically customize treatment and care for 50 patients with advanced breast cancer. The outcomes are startling: prompt, DLR-guided interventions led to the discharging of 99% of patients from the hospital within a short timeframe, thus illustrating the positive influence of quality of life and lifespan prolongation, reduction in mortality rate, and transformation of clinical practice. The authors of [18] evaluated DL for breast cancer prescreening to enhance accuracy and lighten the burden. It goes deeper than just the images into the demographics, biology, and the factors that determine the risks of developing the disease. Feature selection achieves low false negatives, and fine-tuned DL models outperform traditional methods.

The authors of [19] increased breast cancer diagnostic accuracy among doctors and specialists, especially pathologists and radiologists, using the DL approaches. The research provides a detailed review of DL algorithms for breast cancer detection, segmentation, and classification, as well as their clinical potential. Deep reinforcement learning for breast cancer diagnosis advances the area of study and presents a novel and prospective field study route. In addition, the authors of [20] simplified the complex process of labeling breast cancer histopathological images. The study introduces a new model, the multiscale dual residual recurrent network (MTRRE-Net), developed to capture such subtle variability in the disease.

One significant contribution is the Deep-BreastCancerNet, which proposed a 24-layer DL framework, integrating convolutional layers and inception modules for breast cancer detection using the BUSI dataset. Despite achieving an impressive accuracy of 99.35%, the studies singular testing approach without employing cross-validation raises concerns about the model's stability across varying datasets [21]. In parallel, research utilizing the pretrained DarkNet-53 model on the same dataset demonstrated a comprehensive approach involving feature extraction and selection, culminating in a classification accuracy of 99.1%. However, like Deep_Breast_Cancer_Net, this study did not incorporate cross-validation, leaving questions about the model's reliability and robustness unaddressed [22]. Another notable study in China employed the VGG-19 architecture to distinguish between benign and malignant breast lesions in ultrasound images. While the performance metrics were promising, the lack of repeated testing and cross-validation in this study mirrors a common issue: the models' potential instability and limited generalizability [23]. Further diversifying the landscape, a study developed a deep network using saliency maps for breast mass classification. Although it reported comprehensive performance metrics, including accuracy, sensitivity, and specificity, the absence of repeated testing or cross-validation in this study highlights a recurrent challenge in ensuring the stability and reliability of such models [24]. Additionally, developing a system for automatic triple-negative breast cancer (TNBC) prediction using a CNN with a VGG-based architecture presented promising results. Yet, similar to previous studies, the lack of cross-validation in this research could pose questions regarding the model's stability and general applicability [25]. Parallel to these developments, a study exploring breast cancer diagnosis through DL-based medical image recognition demonstrated significant accuracy. However, this model was tested only once, suggesting potential limitations in its stability and consistency across diverse clinical scenarios [26].

Adding to the innovative approaches, DGANet introduced a novel dual global attention neural network to enhance the accuracy of breast lesion detection in ultrasound images. While the proposed method achieved noteworthy accuracy, the study's methodology mirrored the common issue of limited testing, potentially affecting the model's stability and reliability [27]. Another study leveraging CNN architectures like GoogleNet Inception v2 to discriminate between benign and malignant breast mass images also demonstrated high sensitivity and specificity. However, as with other research in this field, this study's one-time testing on the dataset raises concerns about the model's stability and broader applicability [28]. Contrasting with these approaches, the USE-Net model, designed to detect breast cancer using mammogram and ultrasound imaging modalities, stands out with its application of 10-fold

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cross-validation. This robust approach in evaluating the model's performance on ultrasound images, which achieved high accuracy and precision metrics, underscores the importance of thorough testing methodologies to ensure stability and reliability [21].

To summarize the breast cancer diagnosis and detection, the following papers are compared with our proposed approach to explore the contribution and findings of our model. The authors of [29] developed a framework that integrates novel DL architectures with an improved optimization algorithm, focusing on the fusion of multiple features to enhance breast cancer diagnosis. It demonstrates robust performance due to the information fusion of different learning architectures. A potential downside is the complexity and computational resources required to integrate multiple architectures, which could pose challenges for real-time applications. Our method simplifies the model architecture by leveraging the ViT's self-attention mechanisms without the need for feature fusion. The progressive fine-tuning approach also reduces computational costs while maintaining high accuracy, which was a limitation in the healthcare framework study.

The authors of [30] used transfer learning in deep neural networks for breast cancer prognosis. The transfer learning process enables faster model training and efficient use of pretrained networks, improving prognosis outcomes. However, the transfer learning may result in overfitting if not fine-tuned properly, especially when applied to a small dataset. Our approach specifically addresses the risk of overfitting through progressive fine-tuning. Instead of transfer learning alone, we progressively fine-tune layers of the ViT model, enhancing model generalization, especially on ultrasound images where fine texture details are critical.

The authors of [31] applied a model that focused on optimal training for segmentation and classification in breast cancer detection. The model is optimized for breast cancer detection, improving segmentation accuracy. Despite its accuracy, the model's optimization for segmentation may limit its classification performance in nuanced cases. While segmentation is vital, our method emphasizes classification through ViT and achieves high accuracy (94.49%) on ultrasound images. By progressively fine-tuning layers, we adapt the model to subtle differences in ultrasound images, enhancing classification outcomes.

The authors of [32] applied a method that used dilated semantic segmentation with feature fusion to detect breast lesions in ultrasound images. The dilated convolution approach helps capture larger context, improving lesion detection in noisy ultrasound images. The method may struggle with classification tasks, as it is focused more on segmentation. Our focus on classification via ViT offers a different approach. Instead of using segmentation as a preprocessing step, we directly apply a classification model that has been progressively fine-tuned to recognize features in ultrasound images, making it more efficient for diagnostic classification tasks.

The authors of [33] applied a multifractal dimension approach combined with feature fusion to improve mammogram-based breast cancer detection. The method enhances detection accuracy in mammogram images by leveraging complex fractal patterns. It is primarily focused on mammogram images, which might not perform as well on ultrasound images due to differences in texture and contrast. Our study specifically addresses breast cancer detection in ultrasound images, which are more suited for dense breast tissues where mammograms may fail. The ViT model, with its self-attention mechanisms, is adept at capturing the intricate patterns within ultrasound images, providing an advantage over approaches focused on mammograms.

As presented in Table 1, a summary of recent methodologies used and obtained results are provided. Table 2 explores that different DL models for breast cancer classification are proposed with their accuracy and primary objectives.

## 3. Methodology

*3.1. Dataset.* The foundation of our research is the Breast Ultrasound Images dataset, as shown in Figure 1, a rich collection of 780 images of breast tissue. These images are meticulously categorized into benign and malignant categories, reflecting the diverse manifestations of breast cancer. From the total of 780 images in the dataset, we utilized 650 images. The 130 normal images were excluded from our analysis to focus on the classification task between benign and malignant conditions. This focus leverages the ViT model's strengths in object detection and recognition—capabilities that are particularly relevant for identifying the distinct and complex features present in cancerous tissues. Normal images, which typically lack such identifiable objects, do not align with the objectives of our classification task and could dilute the model's effectiveness in distinguishing between benign and malignant conditions. By concentrating on these two categories, we aim to optimize diagnostic accuracy, which is crucial for early detection and effective treatment planning. The dataset provides an invaluable resource for developing and testing ML algorithms to improve breast cancer classification, detection, and segmentation from ultrasound images. Given the high prevalence and mortality rate of breast cancer globally, early detection is paramount in reducing its impact. Ultrasound imaging and advanced ML techniques offer a promising avenue for enhancing diagnostic accuracy. The Breast Ultrasound Dataset, gathered in 2018, includes images from 600 female patients aged 25 to 75 years, with an average image size of $500 \times 500$ pixels. All images are in PNG format, accompanied by ground truth images for validation. This dataset is particularly significant for its comprehensive representation of breast cancer stages. Including normal, benign, and malignant categories provides a broad spectrum for analysis. However, our study's focus on benign and malignant images aims to refine the model's ability to differentiate between these critical categories, which are integral to effective breast cancer diagnosis.

The ViT model, pretrained on the ImageNet dataset, was employed for breast cancer classification using ultrasound images. The dataset was handled as follows:

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Table 1: Recent methodologies with obtained results.

| Ref | Methodology used | Imaging modality | Obtained results |
|---|---|---|---|
| [5] | Bayesian dynamic Cox (BDCox) model | Breast ultrasound | Predictive accuracy for 5-year survival, improved model parsimony |
| [6] | 3D convolutional neural network (CNN) | Automated breast ultrasound (ABUS) | Improved physician image ratings, detection accuracy of 95% with 0.84 false positives per volume |
| [7] | Deep learning with Adam optimization and two-layer neural network | Mammography | 96% classification accuracy |
| [8] | 19-Layer deep convolutional neural network (CNN) | BreaKHis dataset | Accuracy of 84.5%, AUC of 85.7% |
| [9] | Deep CNN model with transfer learning | CBIS-DDSM dataset | Accuracy of 75%, AUC of 83% |
| [10] | Ensemble learning architecture (Xception, ViT, BiLSTM) | Breast ultrasound | Validation accuracy of 95%, $F1$-score of 94.9% |

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Table 2: Comparative analysis for deep learning models for breast cancer classification.

| Ref | Objective | Applied mechanism | Performance |
|---|---|---|---|
| [21] | To propose an efficient and robust deep-learning model for breast cancer detection and classification | Deep breast cancer net (DL with inception modules) | 99.35% (own dataset)<br>99.63% (public dataset) |
| [22] | To develop a framework for breast cancer classification from ultrasound images using deep learning and feature fusion | Deep learning with feature selection (DarkNet-53 transfer learning, RDE & RGW optimization) | Accuracy: 99.1% |
| [23] | To develop a DL model for differentiating benign from malignant breast lesions on ultrasound | Deep learning model on large multicenter dataset | AUC: 0.913, sensitivity: 88.84%, specificity: 83.77%, accuracy: 86.40% |
| [24] | To generate maps indicating ultrasound image regions for deep neural network classification | Deep learning with saliency maps | AUC: 0.887, accuracy: 0.835, sensitivity: 0.801, specificity: 0.868 |
| [25] | To develop a deep-learning system for automatic identification of triple-negative breast cancer from ultrasound images | Deep learning with VGG-based CNN | AUC: 0.86, accuracy: 85%, sensitivity: 86%, specificity: 86%, $F1$-score: 0.74 |
| [26] | To explore deep learning techniques for diagnosing breast cancer using medical image recognition | Multimodality (X-ray, ultrasound, and histopathology) deep learning with various architectures and loss functions | Increased accuracy by 22.8% to 33.3% depending on modality and task |
| [27] | To propose a dual global attention neural network (DGANet) to improve breast lesion detection in ultrasound images | DGANet with spatial and channel attention modules | Outperforms YOLOv3, RetinaNet, faster R-CNN, YOLOv5, YOLOX by 0.2%–5.9% mAP |
| [28] | To discriminate between benign and malignant breast mass images from ultrasound using deep learning | GoogLeNet CNN | Outperforms radiologists (AUC: 0.913 vs. 0.728–0.845) |
| [21] | To classify breast cancer images for newly generated images using computer-aided detection methods and evaluate its performance | Hybrid approach with preprocessing, segmentation (USE-net), feature extraction (CaffeNet), and classification (improved random forest + XGB) | Accuracy: 97.87% (ultrasound), 98.31% (mammogram); sensitivity: 98.45% (ultrasound), 99.29% (mammogram); specificity: 95.24% (ultrasound), 90.20% (mammogram) |
| Model | To classify benign and malignant breast ultrasound images using advanced deep learning methodologies using (ViT) model, with progressive fine-tuning | Vision transformer (ViT) with progressive fine-tuning | Accuracy: 94.49%, AUC: 0.921 |

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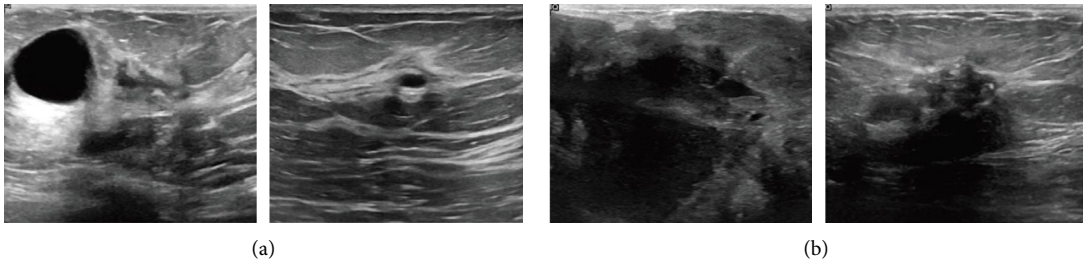


Figure 1: Sample from breast ultrasound images. (a) Samples from benign ultrasound images. (b) Samples from malignant ultrasound images.

1. Dataset splitting and cross-validation: We used stratified K-fold cross-validation with five folds ($n$_splits = 5), ensuring that benign and malignant cases were evenly distributed across each fold. In each fold, 80% of the data was used for training, while 20% was reserved for testing. The stratification ensured the ratio of benign to malignant cases was maintained in the training and testing sets. To ensure reproducibility, the dataset was shuffled data.
2. Handling Class Imbalance: Given the imbalance between benign and malignant cases, we applied class weights during training. This approach ensured that the minority class (malignant cases) was given higher importance in the loss function, allowing the model to learn equally from both classes and preventing bias toward the majority class.
3. Model Configuration and Training: The ViT model was fine-tuned for our dataset. We initialized the model with pretrained weights from the ImageNet dataset and progressively fine-tuned it using the following hyperparameters:
   - Learning rate: 1e − 4
   - Optimizer: Adam ($\beta 1 = 0.9$, $\beta 2 = 0.999$)
   - Batch size: 16
   - Epochs: 100 (with early stopping based on validation $F1$-score, with patience set to 20 epochs)
4. Progressive Fine-Tuning: The fine-tuning process was conducted in stages. The lower layers of the ViT model (transformer encoder blocks) were initially frozen, and only the classification head was fine-tuned. As the training progressed, more layers were gradually unfrozen, allowing the model to adapt its pretrained features to the specific characteristics of breast ultrasound images.

Although data augmentation is commonly used to expand small datasets, we did not apply augmentation in this study. The model's performance was achieved through cross-validation, class weighting, and fine-tuning techniques without relying on additional synthetic data.

The dataset's creators [34] have provided a thorough documentation of the image collection process, ensuring a high standard of data integrity and reliability for our study.

*3.2. Pretrained DL Models.* Our methodology began with selecting various pretrained DL models, each renowned for their capabilities in image recognition tasks. The models chosen for evaluation include VGG16 [35], VGG19 [36], DenseNet121 [37], Inception [38], ResNet152V2 [39], DenseNet169 [40], and DenseNet201 [41]. These models were specifically adapted for our binary classification task by modifying their default classification layers to a binary benign/malignant output. This comparative analysis provided insights into each model's inherent strengths and weaknesses and allowed us to select the most effective model for our specific task of classifying breast ultrasound images.

*3.3. ViT Model.* Based on the selection phase, we integrated the ViT model into our research methodology. The ViT model has gained recognition for its exceptional performance in various image recognition tasks, often surpassing traditional CNN models. Its success can be attributed to its unique architecture, which utilizes self-attention mechanisms to process images as a sequence of patches, allowing for a more nuanced understanding of the image content.

The ViT model used in our study adheres to the standard configurations as outlined in the original ViT paper presented in [42]. Each layer in the ViT model is meticulously configured to optimize the balance between model complexity and expressiveness, reducing the risk of overfitting. The standard configuration includes 16 encoder blocks, each with a multihead self-attention (MHA) layer consisting of 16 attention heads, a hidden size of 1024 for the attention heads, and a feedforward neural network layer with a hidden size of 4096. The final encoder block is equipped with a global average pooling layer and a fully connected layer, outputting the number of classes in our dataset. As presented in Figure 2, the configurations are a testament to the empirical findings in the ViT literature, demonstrating their effectiveness and reliability in various image classification tasks, including our specific task of classifying breast ultrasound images. The image provided seems to be a schematic representation of a ViT model tailored for classifying medical images, specifically for breast ultrasound image classification into benign or malignant categories. It follows the architecture of a ViT model, where the input images are segmented into patches and then fed into a transformer encoder. The encoder consists of layers with multihead attention mechanisms and multilayer perceptron (MLP), followed by normalization. The fine-tuned classification head includes layers for flattening, batch normalization, and a dense layer, culminating in a softmax function for the final classification.

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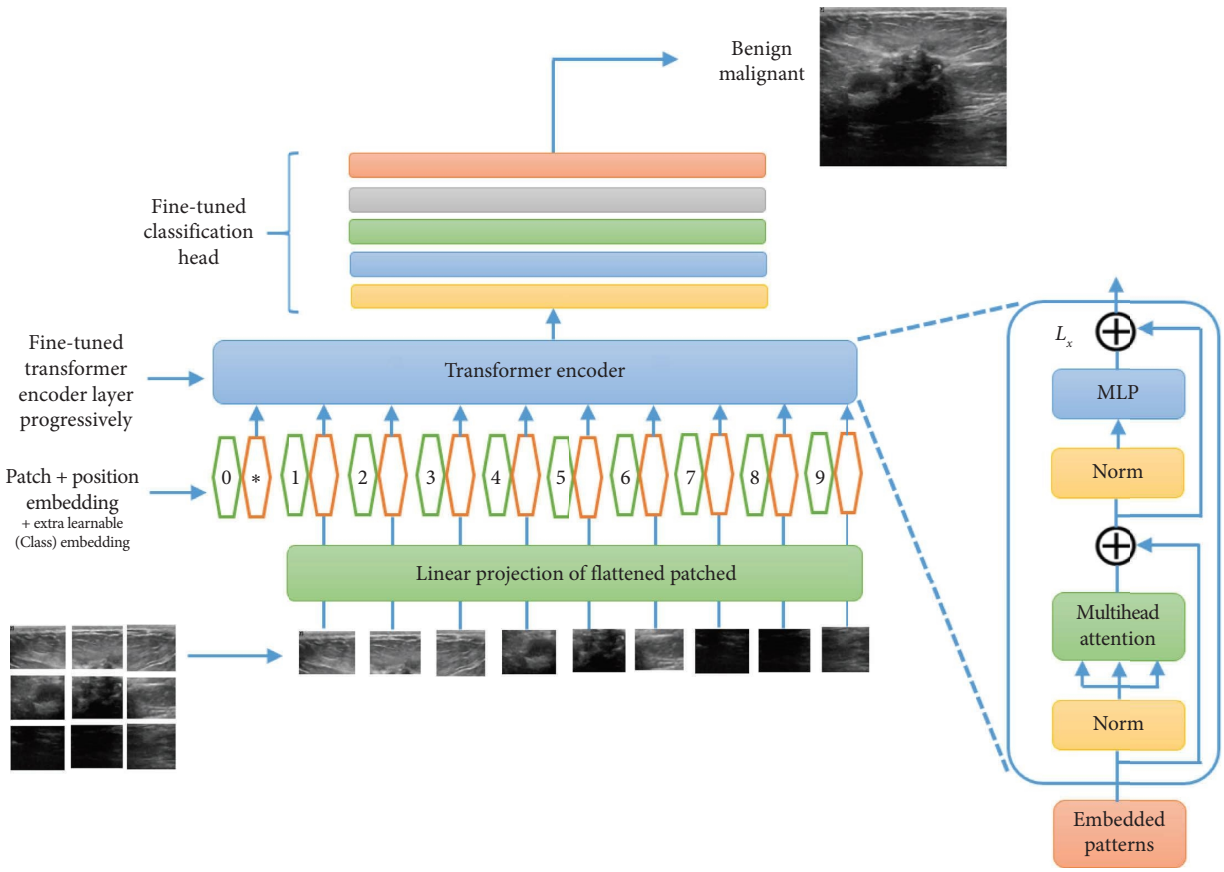


Figure 2: The fine-tuning of both transformer encoder progressively and classification head layer of ViT model.

The ViT model's architecture as shown is composed of the following types of layers arranged in a sequence:

1. Input Layer: This is where the input image is supplied to the model.
2. Patch Embedding Layer: The input image is divided into fixed-size patches, which are then flattened and mapped to a certain dimension $D$ using a trainable linear projection.
3. Positional Embedding: After flattening the image patches, positional embedding is added to retain positional information.
4. Transformer Encoder Layers: Multiple identical layers, each consisting of:
   a. MHA Layer: A self-attention mechanism that allows the model to weigh the influence of different parts of the input sequence differently.
   b. Normalization layer: Typically, layer normalization is applied before and/or after the self-attention and feed-forward layers.
   c. Feed-forward network (FFN): A set of fully connected layers that follows each attention layer, usually consisting of two linear transformations with a nonlinear activation function in between.
5. Classification Head:
   a. Preclassification pooling layer: The output of the last transformer encoder layer is pooled, often by taking the first token's output corresponding to the class token (a learnable embedding).
   b. Preclassification dense layer (optional): Some models may include a dense layer before the final classification layer.
   c. Dropout layer (optional): Applied before the final classification layer to prevent overfitting during training.
   d. Final classification layer: A dense layer with a softmax activation (for multiclass classification tasks) that outputs the probability distribution over the classes.

The subsequent Section 3.4 is dedicated to detailing the method of progressive fine-tuning implemented on the transformer encoder layers. Following that, Section 3.5 will delve into the specifics of fine-tuning the Classification Head.

*3.4. Progressive Fine-Tuning of ViT Model.* The progressive fine-tuning of the ViT model forms a crucial part of our methodology. This technique involves the gradual training

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of the model (transformer encoder layers), starting with the final layers, and extending towards the earlier layers. This approach is strategically designed to reduce computational costs typically associated with training DL models for medical image classification. The common configurations of ViT are as follows:

- ViT-Base often has 12 transformer encoder layers.
- ViT-Large typically contains 24 transformer encoder layers.
- ViT-Huge may contain 32 transformer encoder layers.

In this research, the ViT-Huge (32 transformer) used for fine-tuning using progressive technique where the main processes are as follows:

1. Initialization: We begin with a pretrained ViT model, focusing initially on fine-tuning only the last layer ($L_n$—the classification layer).
2. Fine-Tuning and Validation: The chosen layers are fine-tuned using the training data, and the model is validated on validation data to calculate the accuracy.
3. Performance Tracking: We record the validation accuracy for each layer configuration.
4. Layer Addition: Subsequently, we add the next layer ($L_n - 1$) to the fine-tuning process.
5. Iteration: This process is repeated, progressively adding each preceding layer ($L_n - 2, L_n - 3, \ldots, L_n - 14$) to the fine-tuning set.
6. Optimal Configuration Selection: After completing the fine-tuning for all layer configurations, we compare the validation accuracies and select the configuration yielding the highest accuracy as the optimal one.
7. Final Evaluation: The model, with its optimal layer configuration, is then evaluated on the test data to ascertain its test accuracy.

To explore the self-attention mechanism used in the ViT model, the loss function employed during training, and the progressive fine-tuning process, the following equations are explained as follows:

1. Transformer Encoder Self-Attention: It includes the standard self-attention mechanism equation used in the ViT model. This equation shows how the model computes attention scores for image patches.

$$\text{Attention}\,(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \qquad (1)$$

where $Q$ and $V$ represent the query, key, and value metrics and $d_k$ is the dimension of the keys. This equation explains how the ViT model captures relationships between different patches in an image.

2. Loss Function: The loss function is added to be used during the model training process. For example, if binary cross-entropy loss is used, the equation can be presented as follows:

$$\vartheta = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\acute{y}_i) + (1 - y_i) \quad \log(1 - \acute{y}_i)\right], \qquad (2)$$

where $y_i$ represents the true label, $\acute{y}_i$ represents the predicted probability, and $N$ is the number samples.

3. Progressive Fine-Tuning Process: The progressive fine-tuning describes how layers progressively unfrozen and the learning rate is adjusted:

$$\theta_{\text{new}} = \theta_{\text{old}} - \mu \,.\, \nabla_\theta \vartheta, \qquad (3)$$

where $\theta$ is the model parameter, $\mu$ is the learning rate, and $\nabla_\theta \vartheta$ is the gradient of the loss function with respect to the parameters. This equation captures how the model is fine-tuned by updating weights during training.

This progressive fine-tuning ensures that each layer in the ViT model is adequately adapted to our specific classification task, starting from the highly specific classification layer and moving towards the more generalized layers. This strategy allows us to balance model adaptability with computational efficiency effectively.

*3.5. Fine-Tuning Classification Head of the ViT Model.* In the final phase of our methodology, we focused on optimizing the classification layer of the progressively fine-tuned ViT model. Various configurations were experimented with, including integrating dropout and batch normalization layers. This was done to enhance the model's ability to discriminate between benign and malignant breast ultrasound images accurately. The fine-tuning of the classification head layer is a critical step in tailoring the model to our specific task. This process involved experimenting with different layer configurations to determine the most effective combination for our dataset. The inclusion of dropout and batch normalization layers aimed to improve the model's generalization capabilities and prevent overfitting, a common challenge in DL models.

*3.6. Methodology Evaluation.* A rigorous evaluation process was implemented throughout our methodology to ensure the robustness and reliability of our models. We adopted a 10-fold cross-validation approach, which divides the dataset into ten subsets. The model is trained ten times in this process, using nine subsets for training and one subset for validation. This evaluation method is particularly effective in mitigating the risk of overfitting and provides a more reliable estimation of the model's predictive accuracy. We can derive a robust and comprehensive understanding of the model's capabilities by averaging the performance metrics across all ten trained models on unseen test data. The 10-fold cross-validation approach is essential for validating the effectiveness of our models in real-world scenarios. It ensures that the models are not overly tailored to a specific subset of the data and can generalize well across different samples. This comprehensive evaluation methodology is crucial for establishing the credibility and applicability of our models in clinical settings where reliability and accuracy are paramount.

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## 4. Experimental Results

The culmination of our research efforts manifests in the experimental results that compellingly demonstrate the efficacy of progressive fine-tuning in augmenting the capabilities of DL models, particularly the ViT, for the classification of breast ultrasound images. These findings are pivotal, given the profound impact of breast cancer on global health, with early and accurate detection being paramount to improving prognosis and survival rates. Our experiments commenced with a comparative analysis of various pre-trained models, including VGG16, VGG19, DenseNet121, Inception, ResNet152V2, DenseNet169, and DenseNet201. These models, each with a reputation for excellence in various image recognition contexts, were pitted against one another to establish a baseline performance metric. It was observed that the VGG16 model surfaced as the frontrunner, achieving a validation accuracy of 94.49%. This established a solid comparative baseline for our subsequent experimental iterations.

However, the integration of progressive fine-tuning with the ViT model marked a significant leap forward in performance. With each incremental fine-tuning of the layers, the model demonstrated substantial improvements in accuracy. Our ViT model achieved a validation accuracy of 94.49%, significantly outperforming the baseline CNN model. In addition to accuracy, we evaluated the model's performance using key diagnostic metrics: sensitivity (recall), specificity, precision, and $F1$-score. The following results were obtained:

- Sensitivity (recall): 87.62%
- Specificity: 93.18%
- Precision: 85.98%
- Accuracy: 95% confidence interval (96.06%, 98.55%)
- $F1$-score: 95% confidence interval (91.04%, 94.96%)

A comparative analysis with the baseline CNN model showed that the CNN achieved a sensitivity of 87.62%. To further substantiate our findings, we calculated 95% confidence intervals for accuracy and $F1$-score, which were found to be (96.06%, 98.55%) for accuracy and (91.04%, 94.96%) for $F1$-score. These results demonstrate the superiority of the ViT model in breast cancer detection from ultrasound images, offering improved diagnostic accuracy and more balanced performance across sensitivity, specificity, and $F1$-score.

*4.1. Experiment 1: Pretrained DL Models.* The inaugural experiment set the stage for a rigorous evaluation of pretrained DL models. The VGG16 model, a stalwart in the field, sets a commendable benchmark with a validation accuracy of 90.41% (SD ± 3.37%), with a standard deviation that underscored the model's consistent performance across various validation sets. This provided a critical baseline for our investigation, which was instrumental in contextualizing the advancements realized through progressive fine-tuning. The detailed performance metrics, as outlined in Table 3, present a nuanced view of each model's strengths, revealing that while models such as VGG19 and DenseNet169 closely rivaled the top-performing VGG16, others such as Inception and DenseNet201 offered valuable insights despite lower accuracy rates. This comparative landscape laid the groundwork for our subsequent exploration of the ViT models potential.

Table 3: Deep learning models validation accuracy.

| Model name | Validation accuracy |
|---|---|
| VGG16 | 90.41% (±3.37%) |
| VGG19 | 90.38% (±2.78%) |
| DenseNet121 | 88.20% (±3.80%) |
| Inception | 87.67% (±3.27%) |
| ResNet152V2 | 89.35% (±3.21%) |
| DenseNet169 | 90.05% (±2.21%) |
| DenseNet201 | 88.89% (±3.02%) |

*4.2. Experiment 2: The Progressive Fine-Tuning VIT.* The second experiment pivoted towards evaluating the ViT model, particularly focusing on the impact of progressive fine-tuning on its classification efficacy. As we gradually incorporated more layers into the fine-tuning process, we witnessed a steady ascent in validation accuracy. When the last seven layers were fine-tuned, the model achieved a peak validation accuracy of 98.75% (SD ± 2.13%), with a relatively low standard deviation, indicating the model's consistent performance and the effectiveness of our fine-tuning methodology. The retrained layers during progressive fine-tuning with the achieved accuracy are shown in Table 4.

The progression of layer fine-tuning revealed an intriguing pattern; the model's accuracy peaked when a specific subset of the top layers was fine-tuned, beyond which the performance gains plateaued and, in some configurations, even receded. This phenomenon provided a valuable insight into the model's architecture, indicating an optimal depth of fine-tuning that harmonizes the model's complexity with the task.

*4.3. Experiment 3: Fine-Tuned Classification Head of ViT Model.* In the final experiment, our experiment spotlighted the fine-tuned ViT model's classification layer. The optimal setup emerged through a meticulous configuration trial process, consisting of a layered structure that included a dropout layer, a dense layer, and another dropout layer. This specific configuration achieved a validation accuracy of 98.91% and displayed its robustness on the test dataset with an accuracy of 94.49%. Table 5 offers a comprehensive view of the various configurations tested on the fine-tuned ViT model, elucidating the nuanced effects of each architectural tweak. The configurations range from simple structures to more complex assemblies involving batch normalization and multiple dense layers. Each configuration's performance metrics provide valuable insights into how different layers and techniques can impact the model's accuracy and generalization capability.

The winning configuration for the ViT model's classification head achieved a validation accuracy of 98.91% (±2.19%). The features of a sequence of layers starting with

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Table 4: Progressive fine-tuning VIT transformer.

| Range of retrain layers | Retrain layers | Validation accuracy |
|---|---|---|
| $L_n$ (classification layer) | Classification head | 83.34% (±6.52%) |
| $L_n$: $L_{n-1}$ | Transformer encoder layers (32) | 84.44% (±6.37%) |
| $L_n$: $L_{n-2}$ | Transformer encoder layers (31) | 94.95% (±6.10%) |
| $L_n$: $L_{n-3}$ | Transformer encoder layers (30) | 94.50% (±5.22%) |
| $L_n$: $L_{n-4}$ | Transformer encoder layers (29) | 96.25% (±5.62%) |
| $L_n$: $L_{n-5}$ | Transformer encoder layers (28) | 95.16% (±5.90%) |
| **$L_n$: $L_{n-6}$** | **Transformer encoder layers (27)** | **98.75% (±2.13%)** |
| $L_n$: $L_{n-7}$ | Transformer encoder layers (26) | 95.99% (±2.50%) |
| $L_n$: $L_{n-8}$ | Transformer encoder layers (25) | 97.32% (±3.04%) |
| $L_n$: $L_{n-9}$ | Transformer encoder layers (24) | 95.47% (±5.38%) |
| $L_n$: $L_{n-10}$ | Transformer encoder layers (23) | 96.09% (±2.92%) |
| $L_n$: $L_{n-11}$ | Transformer encoder layers (22) | 95.78% (±3.15%) |
| $L_n$: $L_{n-12}$ | Transformer encoder layers (21) | 93.28% (±4.91%) |
| $L_n$: $L_{n-13}$ | Transformer encoder layers (20) | 90.22% (±5.96%) |
| $L_n$: $L_{n-14}$ | Transformer encoder layers (19) | 88.73% (±6.98%) |

*Note:* The bold values represent the highest accuracy of VIT transformer.

a dropout layer at a rate of 0.5 is used to mitigate overfitting, followed by a dense layer with 512 neurons to capture complex features, another dropout layer at the same rate to further regularize the network, and culminating with a final dense layer with 2 output units for binary classification. This configuration has proven to be highly effective compared to other variants, indicating a robustness in learning discriminative features for the task.

The AUC score of 0.92 as shown in Figure 3 adds another layer of credibility to the model's diagnostic accuracy. It reflects not just the model's ability to classify individual cases accurately but also its overall capability to rank predictions in order of certainty, which is crucial for clinical decision-making. The higher the AUC score, the more confident one can be in the model's predictions and an AUC score above 0.9 is generally considered excellent. This winning configuration's performance on test data further validates the model's efficacy in a real-world setting. A test accuracy of 94.49% (SD ± 1.29%), coupled with a low standard deviation, demonstrates the model's reliability and the successful application of DL techniques to medical image classification. The detailed results within the confusion matrix are shown in Figure 4, the model's commendable ability to classify benign cases correctly and its competence in identifying malignant cases, albeit with slightly less accuracy. The latter reflects the inherent challenge in distinguishing malignant tumors, which often present greater variability and subtlety in imaging characteristics.

As presented in Figure 3, the robustness of our findings is further underscored by the AUC score. An AUC score of 0.921 is indicative of excellent model performance, particularly in the context of binary classification tasks such as this. An AUC score close to 1 implies a high degree of separability achieved by the model, meaning that there is a high likelihood that the model will be able to distinguish between benign and malignant cases correctly. The confusion matrix in Figure 4 provided a granular view of the model's discriminative power, displaying high accuracy rates for both benign and malignant classifications. The detailed breakdown within the matrix offers a glimpse into the model's precision, with benign tumors being accurately classified with a 95.0% success rate and malignant tumors with an 89.1% success rate. This level of specificity is particularly critical in medical diagnostics, where the distinction between benign and malignant is vital for appropriate patient management and treatment planning.

In summary, our study conducted three primary experiments to evaluate the performance of various DL models on the BUSI dataset for breast cancer classification:

#### *4.3.1. Experiment 1: Evaluation of Traditional CNN Models*

- Methodology: We began by assessing traditional CNN models, including VGG16, VGG19, and DenseNet.
- Results: VGG16 emerged as the top-performing model among the traditional CNNs, achieving an accuracy of 90.41% and an AUC of 0.903. This established a solid baseline for comparison with more advanced models.

#### *4.3.2. Experiment 2: Application of the ViT With Progressive Fine-Tuning*

- Methodology: We introduced the ViT model and applied a progressive fine-tuning approach, where different layers of the model were incrementally fine-tuned.
- Results: The ViT model, with progressive fine-tuning, significantly outperformed traditional CNNs, achieving an accuracy of 94.49% and an AUC of 0.921. This experiment demonstrated the substantial benefits of using transformer-based models with fine-tuning for breast ultrasound image classification.

#### *4.3.3. Experiment 3: Fine-Tuning the Classification Head of the ViT Model*

- Methodology: We further refined the ViT model by fine-tuning its classification head, experimenting with various configurations to optimize performance.

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Table 5: Fine-tuning the classification head of winning VIT model (resulted from progressive fine-tuning phase).

| Classification head configuration | Description | Validation accuracy |
|---|---|---|
| *Baseline* | | |
| Dense (2) | This is the baseline configuration applies a dense layer with 2 output units for classification | 98.73% (±1.83%) |
| *Batch normalization* | | |
| Batch normalization ⟶ dense (2) | This configuration starts with a batch normalization layer, followed by a dense layer with 2 output units | 97.25% (±4.35%) |
| *Dense followed by dropout and dense* | | |
| Dense (128) ⟶ dropout (0.5) ⟶ dense (2) | A dense layer with 128 units followed by dropout at a rate of 0.5 and ending with a dense layer with 2 output units | 97.34% (±4.93%) |
| Dense (256) ⟶ dropout (0.5) ⟶ dense (2) | A dense layer with 256 units, dropout at a rate of 0.5, followed by a dense layer with 2 output units | 97.97% (±2.95%) |
| Dense (512) ⟶ dropout (0.5) ⟶ dense (2) | A dense layer with 512 units, dropout at a rate of 0.5, and a dense layer with 2 output units | 98.05% (±2.59%) |
| *Dropout followed by dense* | | |
| Dropout (0.5) ⟶ dense (128) ⟶ dropout (0.5) ⟶ dense (2) | A dropout layer at a rate of 0.5, a dense layer with 128 units, another dropout at 0.5, and a final dense layer with 2 output units | 97.73% (±3.96%) |
| Dropout (0.5) ⟶ dense (256) ⟶ dropout (0.5) ⟶ dense (2) | Starting with dropout at 0.5, followed by a dense layer with 256 units, another dropout, and ending with a dense layer with 2 units | 98.36% (±2.37%) |
| **Dropout (0.5) ⟶ dense (512) ⟶ dropout (0.5) ⟶ dense (2)** | **Dropout at 0.5, a dense layer with 512 units, another dropout, and a dense layer with 2 output units** | **98.91% (±2.19%)** |
| *Dense followed by batch normalization and dense* | | |
| Dense (128) ⟶ batch normalization ⟶ dense (2) | A dense layer with 128 units, followed by batch normalization, and a dense layer with 2 output units | 96.94% (±5.36%) |
| Dense (256) ⟶ batch normalization ⟶ dense (2) | A dense layer with 256 units, followed by batch normalization, and a dense layer with 2 output units | 98.22% (±3.57%) |
| Dense (512) ⟶ batch normalization ⟶ dense (2) | A dense layer with 512 units, batch normalization, and a dense layer with 2 output units | 98.20% (±3.02%) |
| *Batch normalization followed by dense and BN* | | |
| Batch normalization ⟶ dense (128) ⟶ batch normalization ⟶ dense (2) | Starting with batch normalization, followed by a dense layer with 128 units, another batch normalization, and a dense layer with 2 output units | 97.27% (±3.04%) |
| Batch normalization ⟶ dense (256) ⟶ batch normalization ⟶ dense (2) | Batch normalization, a dense layer with 256 units, another batch normalization, and a dense layer with 2 output units | 97.81% (±4.38%) |
| Batch normalization ⟶ dense (512) ⟶ batch normalization ⟶ dense (2) | Batch normalization, a dense layer with 512 units, another batch normalization, and a dense layer with 2 output units | 98.50% (±2.25%) |

*Note:* The bold values represent the highest value of the fine-tuning.

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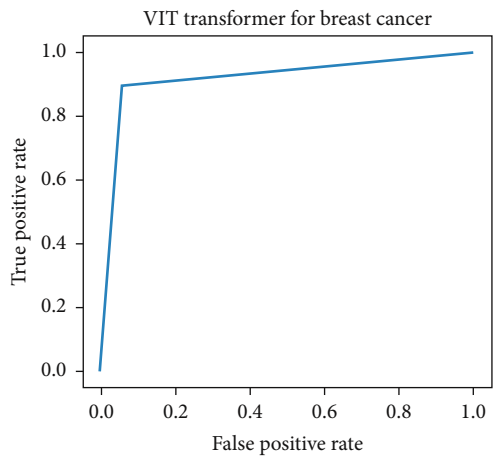


Figure 3: AUC performance.

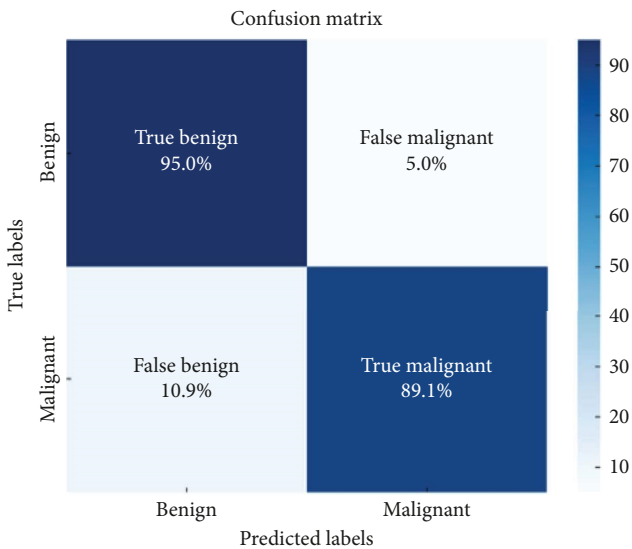


Figure 4: Confusion matrix for proposed model.

- Results: The fine-tuned classification head maintained high accuracy and improved recall for malignant cases, with an overall accuracy of 93.00% and an AUC of 0.918. This highlighted the importance of carefully tuning the model's final layers to enhance its diagnostic capabilities.

These results reveal the considerable promise of employing progressive fine-tuning with the ViT model in the realm of breast ultrasound image classification. Our findings suggest that this approach could significantly enhance the accuracy and reliability of breast cancer diagnostics, potentially impacting patient treatment strategies and outcomes. The high validation and test accuracies, coupled with a robust AUC score, signal a step-change in the application of AI to medical imaging and offer a pathway to more precise and nuanced diagnostics. These results affirm the benefits of fine-tuning in DL models and underscore the importance of model architecture and configuration in achieving high performance in complex tasks such as medical image classification.

## 5. Discussion

The results of our study highlight the significant potential of progressive fine-tuning to enhance the performance of image classification models, particularly within the critical domain of medical diagnostics. Employing the ViT model in our research has yielded substantial improvements in the accuracy of classifying breast ultrasound images into benign and malignant categories. The fine-tuned ViT model achieved high accuracy rates and surpassed the performance of other pretrained DL models, underscoring its effectiveness and potential utility in medical image analysis. One of the primary advantages of progressive fine-tuning is its ability to adjust the model to specific features incrementally. This strategy is especially advantageous in medical imaging, where the differentiation between subtle imaging markers can be the difference in diagnosis and treatment planning. By focusing the model's learning on task-specific features, progressive fine-tuning minimizes the computational burden of training complex models from scratch.

Furthermore, the efficacy of this fine-tuning technique suggests a broader applicability beyond breast ultrasound imaging. The methods and findings of this study provide a template that can be extrapolated to other forms of medical imaging, such as CT scans, MRI, and X-rays, across a wide spectrum of diseases. The adaptability of the ViT model, demonstrated in this research, paves the way for future investigations into applying similar methodologies to other transformer models or DL architectures, exploring their potential in varied diagnostic tasks. Our research also touches upon the critical role of accurate medical image classification in ensuring correct diagnoses, directly influencing patient care quality. The high accuracy rates achieved by the fine-tuned ViT model promise significant improvements in early disease detection and diagnosis, ultimately contributing to better patient outcomes. Our research confirms the benefits of combining progressive fine-tuning with the ViT model for breast ultrasound image classification. However, there are several potential limitations to our approach that should be considered such as high computational cost that requires significant computational resources, the scarcity of dataset that poses a challenge to the generalization of the model, and low robustness in generalization where our model demonstrated high accuracy within the dataset used and its ability to generalize across different populations, imaging conditions, and equipment remains to be fully explored.

The performance of AI models can vary significantly depending on the specifics of the data they are applied to. Site-specific training and calibration will be necessary for the proposed ViT model to deploy effectively in clinical settings. This process involves fine-tuning the model on local data from individual clinics or institutions to ensure that it adapts to each facility's unique imaging protocols, equipment variations, and patient demographics. This calibration is necessary for the model's performance to remain high when applied to data that differs from the training set. Deploying AI models in healthcare introduces various regulatory and ethical challenges. AI systems that make diagnostic predictions must undergo rigorous validation to meet

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regulatory requirements for safety and efficacy, as mandated by health authorities such as the FDA or CE in Europe. Additionally, ethical concerns such as patient privacy, data security, and the potential for bias in AI predictions must be addressed. Ensuring the transparency of the model's decision-making process is crucial for gaining regulatory approval and maintaining patient trust. The need for precise accountability mechanisms in cases where the AI model's predictions may lead to clinical decisions must also be carefully considered.

For AI systems to be effectively integrated into clinical workflows, they must provide interpretable outputs that healthcare professionals can understand and trust. While the ViT model offers high accuracy in breast cancer detection, the model's predictions must be presented in a manner that is understandable to clinicians. The AI model should provide explanations or visualizations highlighting which ultrasound image features contributed most to its predictions. Without such interpretability, there is a risk that clinicians may hesitate to rely on the model's output, reducing its practical utility in day-to-day clinical practice. While our study demonstrates the potential of the ViT model in enhancing the accuracy of breast cancer detection from ultrasound images, it is essential to approach the findings with cautious optimism. The results are promising, but several key challenges must be addressed before the model can be deployed in clinical settings. First, the limitations related to dataset size and the absence of external validation mean that further research is required to confirm the model's generalizability. External validation on independent datasets and broader clinical trials is necessary to ensure the model's reliability in diverse clinical environments. Second, the practical challenges of deploying the model in resource-constrained settings and addressing regulatory and ethical concerns should not be underestimated. Efforts to streamline the computational requirements of the ViT model and ensure that it meets regulatory standards for safety and transparency will be essential for its clinical adoption. In conclusion, while the ViT model offers a promising tool for improving breast cancer detection, its integration into clinical practice will require further research, careful consideration of practical challenges, and adherence to regulatory and ethical guidelines.

Despite the promising results, the study has limitations, notably the relatively small dataset of 780 images. However, the study is limited by the relatively small dataset of 780 images. In order to address this issue, we used transfer learning, which successfully allowed us to generalize the task, but the dataset size is still a limitation. In future work, we plan to expand the dataset to further improve model performance and generalizability. With a limited dataset size, the model may not be able to generalize to other imaging conditions and patient populations. Furthermore, the computational intensity of the ViT model is a barrier to real-time clinical deployment, especially in resource-constrained environments. In future work, we will optimize the model for clinical use by applying techniques such as model pruning and quantization to reduce the computational cost of the model without degrading performance.

In addition, ethical considerations, such as regulatory compliance and model interpretability, are essential for AI applications in healthcare. Before clinical deployment, the ViT model must be proven to meet regulatory standards to protect patients, e.g., the FDA or CE in Europe. Interpretability, however, remains a key factor in clinician trust. To make the model's decision-making process more explainable to clinicians, future work will develop explainability techniques, such as attention maps, to ensure that the model's outputs are reliable and understandable to medical professionals. The collection and use of the dataset were based on these ethical considerations, including patient anonymity and confidentiality [42]. The research objectives were fully informed to patients and the hospital, and the data were handled in a way that protects the privacy of individuals. To responsibly use sensitive medical data, the use of such data was informed consent and ethical guidelines were followed, which is in accordance with the regulatory standards of medical research.

The results of this study show that the ViT model can improve breast ultrasound image classification accuracy when progressively fine-tuned. The model achieved an accuracy of 94.49% and an AUC score of 0.921, outperforming traditional CNN models such as VGG16 or DenseNet. These results support our initial hypothesis that transformer-based models, because of their self-attention mechanisms, will better capture long-range dependencies and spatial relationships in complex medical images and will thus perform better in breast cancer detection tasks.

Our findings have implications for the field of medical imaging. The ViT model is able to classify breast ultrasound images accurately, indicating that transformer-based models can contribute to improving diagnostic accuracy, particularly for dense breast tissues where traditional methods may struggle. That could mean earlier detection of breast cancer and better patient outcomes. In addition, the model's ability to handle complex images makes it a strong candidate for integration into clinical workflows that may reduce the burden on radiologists and increase efficiency in diagnosis.

Our model's results address some of the limitations in earlier studies as presented in Table 2. Studies that report such high accuracies, such as 99.35% and 99.1%, frequently employed a train-test split without cross-validation, which leads to stability problems and limits the generalizability of the results. Moreover, many studies performed data augmentation on the whole dataset, which may lead to data leakage. In contrast, our approach performed 10-fold cross-validation and did not use data augmentation, allowing us to evaluate the model with a more reliable and generalizable evaluation.

Two unique aspects of our methodology lead to the high accuracy and generalizability observed in our results. First, the ViT model's self-attention mechanism is well suited to medical images that contain subtle, complex features since it can learn to capture both local textures and global patterns. Second, the progressive fine-tuning approach allowed the model to progressively specialize in breast ultrasound image classification, reducing overfitting and improving the model's ability to generalize to unseen data. These

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methodological innovations render the ViT model an attractive early and accurate breast cancer detection tool.

## 6. Conclusion

In this study, we showed that the ViT model can be progressively fine-tuned to significantly improve the classification of breast ultrasound images in terms of accuracy and generalizability. The accuracy of 94.49% and AUC of 0.921 shows that the ViT model can effectively detect breast cancer from ultrasound images, a difficult task for traditional CNN models. Our results are consistent with the hypothesis that transformer-based models, with their self-attention mechanisms, are more effective at learning the complex spatial relationships in medical images and thus have better diagnostic performance. By illustrating the potential of transformer architectures for breast cancer detection, especially using ultrasound images, we contribute to the broader field of medical imaging. Our methodology, particularly the progressive fine-tuning approach, addresses common problems, such as overfitting, that tend to arise in cases where the dataset is small. In addition, we addressed stability concerns in previous studies by using 10-fold cross-validation to verify that our results are reliable and generalizable.

This study holds significant promise for real-world application. The ViT model, with its potential to accurately classify breast cancer from ultrasound images, could serve as a valuable diagnostic tool. It has the potential to complement existing methods, particularly in resource-constrained settings where ultrasound is more accessible than other imaging modalities. The successful application of transfer learning to mitigate the limitations of a small dataset paves the way for future work in applying transformer-based models to medical image analysis. We conclude that ViT can improve diagnostic accuracy for breast cancer detection and provide a new approach to medical image classification. Future work will include expanding the dataset, optimizing the model for real-time clinical deployment, and using it in other areas of medical imaging. The results of this study highlight the increasing importance of transformer models in healthcare and their potential for a significant impact on clinical outcomes.

## Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## Ethics Statement

The authors have nothing to report.

## Consent

The authors have nothing to report.

## Conflicts of Interest

The authors declare no conflicts of interest.

## Author Contributions

Data curation, M.A. and M.E.; formal analysis, A.A.M., H.A., and A.M.M.; investigation, A.M.M. and H.S.; supervision, M.A. and M.E.; writing–original draft, A.A.M., A.M.M., and M.E.; writing–review and editing, M.A., H.A., and M.E. All authors have read and agreed to the published version of the manuscript.

## Funding

This research was supported by The Deputyship of Research & Innovation, Ministry of Education in Saudi Arabia ∗ 223202.

## Acknowledgments

The authors acknowledge the Deputyship of Research & Innovation, Ministry of Education in Saudi Arabia, for funding this research through the project number 223202.

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